

Chapter 7 Answer Key

Introduction to Statistics — MATH& 146

Section 7.2 Exercises

7.2.1. $n_1 = 40$, $\sigma_1 = 8$; $n_2 = 50$, $\sigma_2 = 6$.

$$\text{SE} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{64}{40} + \frac{36}{50}} = \sqrt{1.600 + 0.720} = \sqrt{2.320} \approx \mathbf{1.523}.$$

7.2.2. If $\mu_1 = \mu_2$, the expected value of $\bar{x}_1 - \bar{x}_2$ is **0**. Under H_0 , the two population means are equal, so their difference is zero, and on average the two sample means should be equal as well. If the observed difference is very far from 0 — in either direction — it is unlikely to have arisen by sampling variation alone when H_0 is true. A large $|\bar{x}_1 - \bar{x}_2|$ is therefore evidence against $H_0 : \mu_1 = \mu_2$.

7.2.3. Because $\bar{x}_1$ and $\bar{x}_2$ are independent random variables, the variance of their difference is the *sum* of their individual variances:

$$\text{Var}(\bar{x}_1 - \bar{x}_2) = \text{Var}(\bar{x}_1) + \text{Var}(\bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Averaging the variances would be wrong: it would undercount the total variability in the difference. Intuitively, two sources of randomness contribute to how far $\bar{x}_1 - \bar{x}_2$ can stray from zero, and both must be accounted for. We take the square root at the end to convert from variance to standard deviation (the standard error).

Section 7.3 Exercises

7.3.1. Fertilizer A: $n_1 = 30$, $\bar{x}_1 = 42.1$, $\sigma_1 = 5.8$. Fertilizer B: $n_2 = 35$, $\bar{x}_2 = 39.4$, $\sigma_2 = 6.2$. $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 > \mu_2$ (right-tailed). $\alpha = 0.05$.

Step 1. Both $n_1, n_2 \geq 30$. ✓

Step 2.

$$\begin{aligned}\text{SE} &= \sqrt{\frac{5.8^2}{30} + \frac{6.2^2}{35}} = \sqrt{\frac{33.64}{30} + \frac{38.44}{35}} = \sqrt{1.1213 + 1.0983} = \sqrt{2.2196} \approx 1.4898. \\ z &= \frac{42.1 - 39.4}{1.4898} = \frac{2.70}{1.4898} \approx \mathbf{1.81}.\end{aligned}$$

Step 3. Right-tailed: $p = P(Z \geq 1.81) = 1 - P(Z \leq 1.81) = 1 - 0.9649 = \mathbf{0.0351}$.

Step 4. $p = 0.0351 < \alpha = 0.05$. **Reject H_0 .** There is sufficient evidence at the 5% level to conclude that Fertilizer A produces a higher mean yield than Fertilizer B.

- 7.3.2.** At $\alpha = 0.01$: $p = 0.0351 > 0.01$. **Fail to reject H_0 .** The conclusion changes. This illustrates that the choice of α is consequential: the same data provide sufficient evidence at the 5% threshold but not at the 1% threshold. A stricter significance level demands stronger evidence before rejecting the null hypothesis. The significance level should be chosen in advance, based on the cost of a false rejection, not adjusted after seeing the p-value.
- 7.3.3.** The two statements $H_0 : \mu_1 = \mu_2$ and $H_0 : \mu_1 - \mu_2 = 0$ are mathematically identical — subtracting μ_2 from both sides of $\mu_1 = \mu_2$ gives $\mu_1 - \mu_2 = 0$. We typically write $H_0 : \mu_1 = \mu_2$ because it is more natural to say the two groups have the same mean than to speak of a difference equaling zero. The test statistic, however, is naturally expressed as $(\bar{x}_1 - \bar{x}_2)/SE$, which makes the “difference = 0” framing explicit in the formula.

Section 7.4 Exercises

- 7.4.1.** Using the fertilizer data: $\bar{x}_1 - \bar{x}_2 = 2.70$, $SE = 1.4898$, $z^* = 1.645$ (90% CI).

$$2.70 \pm 1.645 \times 1.4898 = 2.70 \pm 2.45 \Rightarrow (0.25, 5.15).$$

The interval (0.25, 5.15) does **not** contain 0. This is consistent with rejecting H_0 at $\alpha = 0.05$ (one-tailed): the data suggest Fertilizer A genuinely out-performs Fertilizer B, with the true yield advantage estimated to be between 0.25 and 5.15 bushels per acre. The interval is wide, so while the direction of the effect is clear, its magnitude is uncertain.

- 7.4.2.** A 95% CI for $\mu_1 - \mu_2$ is $(-3.2, 8.6)$.
- (a) Yes, it contains 0 (since $-3.2 < 0 < 8.6$).
 - (b) Because 0 is inside the 95% CI, we **fail to reject $H_0 : \mu_1 = \mu_2$** at $\alpha = 0.05$ (two-tailed). The data are consistent with the two groups having the same mean.
 - (c) The midpoint of the interval is $(-3.2 + 8.6)/2 = 2.7 > 0$, so $\bar{x}_1 - \bar{x}_2 = 2.7 > 0$: **Group 1 has the larger sample mean.** The interval spans both sides of 0, meaning the true difference could favor either group, but our best single estimate is that Group 1 is higher by 2.7 units.

Section 7.5 Exercises

- 7.5.1.** Men: $\hat{p}_1 = 210/400 = 0.525$. Women: $\hat{p}_2 = 195/350 = 0.557$. $H_0 : p_1 = p_2$ vs. $H_a : p_1 \neq p_2$ (two-tailed). $\alpha = 0.05$.

Step 1. Conditions: $400(0.525) = 210 \geq 10$, $400(0.475) = 190 \geq 10$, $350(0.557) = 195 \geq 10$, $350(0.443) = 155 \geq 10$. ✓

Step 2.

$$\hat{p}_{\text{pool}} = \frac{210 + 195}{400 + 350} = \frac{405}{750} = 0.5400.$$

$$SE = \sqrt{0.5400(0.4600) \left(\frac{1}{400} + \frac{1}{350} \right)} = \sqrt{0.2484 \times 0.005357} \approx \sqrt{0.001331} \approx 0.0365.$$

$$z = \frac{0.525 - 0.557}{0.0365} \approx \frac{-0.032}{0.0365} \approx -0.88.$$

Step 3. Two-tailed: $p = 2(1 - P(Z \leq 0.88)) = 2(1 - 0.8106) = 2(0.1894) = \mathbf{0.3788}$.

Step 4. $p = 0.3788 > 0.05$. **Fail to reject H_0 .** There is insufficient evidence to conclude that support for Candidate A differs by gender.

7.5.2. Under H_0 , we assume $p_1 = p_2 = p$ for some common unknown value p . The most efficient estimate of that common p uses information from *both* samples combined — that is $\hat{p}_{\text{pool}}$. Using each group's separate $\hat{p}$ would ignore the constraint that H_0 imposes, wasting information and producing a less accurate standard error. This parallels the one-sample proportion test (Chapter 6), where we used the claimed p_0 in the standard error rather than the observed $\hat{p}$, for the same reason: we build the null distribution assuming H_0 is true.

7.5.3. $\hat{p}_1 = 0.45$ ($n_1 = 200$, $x_1 = 90$); $\hat{p}_2 = 0.38$ ($n_2 = 180$, $x_2 = 68$).

$$\hat{p}_{\text{pool}} = \frac{90 + 68}{200 + 180} = \frac{158}{380} \approx \mathbf{0.4158}.$$

$$\text{SE} = \sqrt{0.4158(0.5842) \left(\frac{1}{200} + \frac{1}{180} \right)} = \sqrt{0.2430 \times 0.01056} \approx \sqrt{0.002565} \approx 0.0506.$$

$$z = \frac{0.45 - 0.38}{0.0506} = \frac{0.07}{0.0506} \approx \mathbf{1.38}.$$

Chapter 7 Review

1. Surgical recovery: A ($n_1 = 55$, $\bar{x}_1 = 7.2$, $\sigma_1 = 2.1$) vs. B ($n_2 = 48$, $\bar{x}_2 = 8.0$, $\sigma_2 = 2.4$).

$$\text{SE} = \sqrt{\frac{2.1^2}{55} + \frac{2.4^2}{48}} = \sqrt{\frac{4.41}{55} + \frac{5.76}{48}} = \sqrt{0.0802 + 0.1200} = \sqrt{0.2002} \approx 0.4474 \text{ days}.$$

(a) $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 < \mu_2$ (left-tailed; Procedure A is expected to have a shorter recovery time). $\alpha = 0.05$.

$$z = \frac{7.2 - 8.0}{0.4474} = \frac{-0.8}{0.4474} \approx -1.79.$$

Left-tailed: $p = P(Z \leq -1.79) = 1 - 0.9633 = \mathbf{0.0367}$.

$p = 0.0367 < 0.05$. **Reject H_0 .** There is sufficient evidence at the 5% level that Procedure A has a shorter mean recovery time than Procedure B.

(b) **95% CI for $\mu_1 - \mu_2$:**

$$(7.2 - 8.0) \pm 1.96 \times 0.4474 = -0.8 \pm 0.877 \Rightarrow \mathbf{(-1.677, 0.077)}.$$

(c) The 95% CI just barely contains 0 (upper bound ≈ 0.077), while the left-tailed test at $\alpha = 0.05$ rejected H_0 . These results are **consistent**, not contradictory. A two-sided 95% CI corresponds to a *two-tailed* test at $\alpha = 0.05$; a one-sided test at $\alpha = 0.05$ is equivalent to checking a 90% two-sided CI. The 90% CI is $-0.8 \pm 1.645(0.4474) = (-1.536, -0.064)$, which *does not* contain 0, confirming the one-tailed rejection.

- (d) The 95% CI $(-1.677, 0.077)$ includes values ranging from Procedure A being 1.7 days faster to Procedure B being 0.1 days faster. While the result is statistically significant (one-tailed), the interval is wide and its upper end is essentially zero. The true advantage of Procedure A could be very small — perhaps less than clinically meaningful. Statistical significance does not guarantee practical significance; additional context (cost, risk, patient preference) is needed to judge whether the difference matters in practice.

2. Office satisfaction: X ($\hat{p}_1 = 68/100 = 0.68$) vs. Y ($\hat{p}_2 = 54/90 = 0.600$).

$H_0 : p_1 = p_2$ vs. $H_a : p_1 \neq p_2$ (two-tailed). $\alpha = 0.05$.

Step 1. Conditions: $100(0.68) = 68$, $100(0.32) = 32$, $90(0.60) = 54$, $90(0.40) = 36$ — all ≥ 10 . ✓

Step 2.

$$\hat{p}_{\text{pool}} = \frac{68 + 54}{100 + 90} = \frac{122}{190} \approx 0.6421.$$

$$\text{SE} = \sqrt{0.6421(0.3579) \left(\frac{1}{100} + \frac{1}{90} \right)} = \sqrt{0.2298 \times 0.02111} \approx \sqrt{0.004851} \approx 0.0697.$$

$$z = \frac{0.68 - 0.600}{0.0697} = \frac{0.080}{0.0697} \approx 1.15.$$

Step 3. Two-tailed: $p = 2(1 - P(Z \leq 1.15)) = 2(1 - 0.8749) = 2(0.1251) = \mathbf{0.2502}$.

Step 4. $p = 0.2502 > 0.05$. **Fail to reject H_0 .** There is insufficient evidence to conclude that job satisfaction rates differ between the two offices. The observed 8-percentage-point gap could plausibly be due to sampling variation.

3. Paired vs. independent:

- (a) **Paired.** The same 40 students provide both a pre-test and a post-test score; each student's two scores are naturally linked. The relevant analysis is the within-student difference (post – pre), not a comparison of two separate groups.
- (b) **Independent.** Residents of two different zip codes are separate, unrelated individuals. No pairing exists between observations in the two groups.
- (c) **Independent.** The two groups consist of entirely different patients; no observation in the Drug A group corresponds to a specific observation in the Drug B group.
- (d) **Paired.** The same 30 athletes are measured twice — at the start and end of the season. Each athlete's two measurements form a natural pair, and the analysis should focus on the within-athlete change in heart rate.

4. The reporter's statement contains two errors. First, a p-value of 0.03 means: *if the null hypothesis were true*, there would be only a 3% probability of observing a difference this large or larger by chance. It does not give the probability that the two groups are "genuinely different" (that would require a Bayesian framework, not a classical hypothesis test). Second, "97% chance" inverts the logic of the p-value: the p-value is a probability about the data given H_0 , not a probability about H_0 given the data. A correct statement: "The two-sample test produced $p = 0.03$. Assuming no true difference between the groups, a result this extreme would occur only 3% of the time by chance. At the 5% significance level, we reject the null hypothesis of equal means and conclude the data provide evidence of a real difference."

5. Model response (writing prompt):

All four z-tests share the same core structure: a test statistic of the form (observed statistic – null value)/standard error, which is compared to the standard normal distribution to produce a p-value that is then compared to α . What changes between them is the statistic being tested (mean or proportion, one sample or two), the null value (μ_0 , p_0 , 0 for differences), and the standard error formula (which depends on whether we are working with means or proportions, and on the sample size or sizes involved). Despite these surface differences, the underlying logic — measure how many standard errors the observed statistic lies from the null claim, and ask how likely that is if H_0 is true — is identical in all four cases.